

LOCALLY SPLIT-RANGE PRE-HILBERT-SCHMIDT OPERATORS FACTOR THROUGH HILBERT-SCHMIDT SPACES

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STATUS

Status: strengthened partial result, likely valid. This proves the final Abdillāh–Esterle–Haak conjecture [1] for pre-Hilbert-Schmidt operators whose natural range factorization is locally split on both sides. It does not solve the general case.

SOURCE PROBLEM

In Section 5 of [1], the authors define $u \in \mathcal{L}(E, F)$ to be pre-Hilbert-Schmidt if $wuv : H_1 \rightarrow H_2$ is Hilbert-Schmidt for every pair of Hilbert spaces H_1, H_2 and every pair of bounded operators $v : H_1 \rightarrow E, w : F \rightarrow H_2$. They denote this class by $\text{PS}_2(E, F)$.

At the end of the section, they conjecture that every pre-Hilbert-Schmidt operator $u : E \rightarrow F$ factors through a Hilbert-Schmidt space. They also prove two facts used here:

$$\text{PS}_2(E, H) = \Pi_2(E, H)$$

for every Banach space E and Hilbert space H , and a Banach space Y is Hilbert-Schmidt iff $\mathcal{L}(Y, H) = \Pi_2(Y, H)$ for every Hilbert space H .

PROOF INTUITION

The source's condition is global in the outer Hilbert-space tests, but the 2-summing condition is finite-dimensional in the input sequence. Thus bounded inverses are not really needed on all of Y . It is enough to extend and lift each finite-dimensional piece of Y with uniform constants. The only extra point is that a pre-Hilbert-Schmidt operator has a uniform testing constant; otherwise a direct sum of bad Hilbert tests would contradict the definition.

UNIFORM TESTING LEMMA

Theorem 1. *Let $u : E \rightarrow F$ be pre-Hilbert-Schmidt. There is a constant $C_u < \infty$ such that, for every Hilbert space H and every bounded $W : F \rightarrow H$, the operator $Wu : E \rightarrow H$ is 2-summing and*

$$\pi_2(Wu) \leq C_u \|W\|.$$

Proof. For fixed W , Proposition 5.4 of [1] gives $Wu \in \Pi_2(E, H)$. It remains to see that the constants are uniform over Hilbert targets. Suppose not. Then for each n there are Hilbert spaces H_n, K_n and contractions $W_n : F \rightarrow H_n, V_n : K_n \rightarrow E$ such that

$$\|W_n u V_n\|_{\text{HS}} > 2^{2^n}.$$

Let $K = (\oplus_n K_n)_2$ and $H = (\oplus_n H_n)_2$. Define

$$V((x_n)) = \sum_{n \geq 1} 2^{-n} V_n x_n, \quad Wf = (2^{-n} W_n f)_{n \geq 1}.$$

Then $V : K \rightarrow E$ and $W : F \rightarrow H$ are bounded. Since u is pre-Hilbert-Schmidt, $WuV : K \rightarrow H$ is Hilbert-Schmidt. But its diagonal compression to the n th summand is $2^{-2n}W_nuV_n$, so

$$\|WuV\|_{\text{HS}}^2 \geq \sum_{n \geq 1} 2^{-4n} \|W_nuV_n\|_{\text{HS}}^2 > \sum_{n \geq 1} 1,$$

a contradiction. $\square$

THEOREM

Theorem 2. *Let E, F, Y be Banach spaces and let $u : E \rightarrow F$ be pre-Hilbert-Schmidt. Suppose that*

$$u = jq, \quad q : E \rightarrow Y, \quad j : Y \rightarrow F,$$

where q is onto and j is an isomorphic embedding. Assume there are constants $\lambda, \mu < \infty$ such that:

- *for every finite-dimensional $M \subset Y$ there is $R_M : M \rightarrow E$ with $qR_M = \text{id}_M$ and $\|R_M\| \leq \lambda$;*
- *for every finite-dimensional $M \subset Y$ there is $L_M : F \rightarrow M$ with $L_M j|_M = \text{id}_M$ and $\|L_M\| \leq \mu$.*

Then Y is a Hilbert-Schmidt space. Consequently u factors through a Hilbert-Schmidt space.

Proof. Let H be a Hilbert space and let $B : Y \rightarrow H$ be bounded. We prove that B is 2-summing. Let $y_1, \dots, y_n \in Y$ and put $M = \text{span}\{y_1, \dots, y_n\}$. Choose R_M and L_M as in the hypotheses, and set

$$W = B|_M L_M : F \rightarrow H.$$

Then $\|W\| \leq \mu \|B\|$. By the uniform testing lemma, $Wu : E \rightarrow H$ is 2-summing with $\pi_2(Wu) \leq C_u \mu \|B\|$. Since $u = jq$ and $L_M j$ is the identity on M , for $1 \leq k \leq n$ we have

$$WuR_M y_k = B L_M j q R_M y_k = B y_k.$$

Therefore

$$\left(\sum_{k=1}^n \|B y_k\|^2 \right)^{1/2} \leq C_u \mu \|B\| \|(R_M y_k)_{k=1}^n\|_{\ell_2^w(E)}.$$

Finally

$$\|(R_M y_k)_{k=1}^n\|_{\ell_2^w(E)} \leq \lambda \|(y_k)_{k=1}^n\|_{\ell_2^w(Y)}.$$

Thus $\pi_2(B) \leq C_u \lambda \mu \|B\|$.

Every bounded operator from Y into an arbitrary Hilbert space is 2-summing. By Theorem 5.6 of [1], Y is a Hilbert-Schmidt space. The displayed factorization $u = jq$ is then a factorization through a Hilbert-Schmidt space. $\square$

Corollary 3. *Let $u : E \rightarrow F$ be pre-Hilbert-Schmidt. If $u = jq$ and q, j have bounded right and left inverses, respectively, then the intermediate space is a Hilbert-Schmidt space.*

Proof. Use the same inverses on every finite-dimensional subspace. $\square$

Corollary 4. *Let $u : E \rightarrow F$ be pre-Hilbert-Schmidt. Suppose $u(E)$ is closed, locally complemented in F with a uniform constant, and the quotient map $E \rightarrow u(E)$ locally lifts finite-dimensional subspaces with a uniform constant. Then $u(E)$ is a Hilbert-Schmidt space, and u factors through it.*

Proof. Take $Y = u(E)$, let $j : Y \hookrightarrow F$ be the inclusion, and apply the theorem. $\square$

Corollary 5. *If $P : E \rightarrow E$ is a pre-Hilbert-Schmidt projection, then $P(E)$ is a Hilbert-Schmidt space.*

Proof. Apply the theorem with $Y = P(E)$, $q = P : E \rightarrow Y$, $j : Y \hookrightarrow E$, $R = j$, and $L = P$. $\square$

LIMITATIONS AND FAILED STRONGER ROUTE

The result needs both local splitting hypotheses. Complemented range alone is not enough for this proof: transferring a 2-summing estimate from $Bq : E \rightarrow H$ to $B : Y \rightarrow H$ through a quotient map $q : E \rightarrow Y$ would require uniform lifting of finite weak ℓ_2 configurations, which is false in general. For instance, a quotient map $\ell_1 \rightarrow \ell_2$ would otherwise force the identity on ℓ_2 to be 2-summing.

Thus the unsolved part of the source conjecture remains the genuinely non-locally-split case, where Hilbert-valued maps on finite pieces of the range cannot be moved back and forth through the factorization with uniform constants.

NOVELTY CHECK

The cheap run indexes were searched for arXiv:1406.7546, the source title, and the terms “pre-Hilbert-Schmidt”, “Hilbert-Schmidt space”, and “factorization”. A web search for exact French and English phrases around the conjecture found the source arXiv paper and the *Studia Mathematica* journal page, but no later exact answer to the full conjecture. This packet is therefore recorded as a strengthened partial result, not a literature answer and not a full solution.

REFERENCES

- [1] Said Amana Abdillah, Jean Esterle, and Bernhard H. Haak, *Sur quelques extensions au cadre Banachique de la notion d'opérateur de Hilbert-Schmidt*, arXiv:1406.7546; *Studia Mathematica* 227 (2015), 193–218. <https://arxiv.org/abs/1406.7546>